

{DALAS} Drug Repurposing Project

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“How can we repurpose drugs for new uses?”

“Comment découvrir de nouvelles utilisations de médicaments existants pour d’autres maladies ?”

The discovery of new drugs and their further approval is -although important- limited and extremely expensive. Therefore, it’s of interest to find new applications for already known and approved (safe) molecules. Public research laboratories and pharmacological giants are currently experimenting on ways to solve this issue using vast amounts of chemical, biological and clinical data. Being able to predict new drug-disease associations helps to narrow down the list of potential candidate molecules allowing to set priorities for new clinical trials.

We plan to concentrate our efforts on a restrained set of diseases - a few dozen autoimmune diseases (e.g. psoriasis, rheumatoid arthritis, lupus) and several hundreds of drugs conventionally used for this type of diseases and optionally including, for broader exploration, a random set of drugs used for other disease groups.

We plan to obtain three types of data. First, **chemical features** of drugs such as molecular weight, molecular fingerprints, and targets (restrained to proteins and/or pathways). Then, **characteristics of diseases** - their classification, textual description, molecules and/or pathways involved in the pathogenesis. And finally **pairwise drug-disease information** such as whether the drug has an approved clinical indication, proved therapeutic efficacy (or inefficacy), as well as co-occurrence in medical papers’ abstracts and scores based on the proximity of their embeddings from LLMs like BioBERT, PubMedBERT. We can access this information by addressing public database APIs - ChEMBL, RDKit, FDA, PubMed, as well as scraping reputable websites such as MedlinePlus and DrugBank. Data cleaning, beyond standard procedures, will first involve handling all possible synonyms of drugs and diseases. We can explore our data by analyzing how drug-disease associations cluster in the chemical feature space using PCA.

We can adapt two approaches to model this data. First, we can set a baseline machine learning algorithm for binary classification of a given drug-disease pair - predict whether this pair is a potential match, i.e., whether the drug is likely to be effective in a clinical trial. This can involve models such as logistic regression, random forest or XGBoost. More advanced graph-based models could be used to predict new associations (edges) between drugs and diseases based on shared targets and pathways.

To evaluate the models, in addition to standard methods such as ROC-AUC and cross-validation, we plan to extract the timestamps of drug indication approvals and use them to test the model on more recent discoveries relative to the training data, avoiding data leakage from future information during cross-validation.